

## **Rótuloselectronicos.net control communication protocol**

### **1、Intruccion**

#### ***General agreement of communication***

Data packet is used to communicate between the PC and the controller, in order to enhance the reliability of data, expanding capabilities to deal with images and other data.

##### **Communication process :**

- a).PC send a data packet to the controller ;
- b).The controller received the data packet, analysis and processing of data packet, and then return a data packet to PC if necessary ;
- c). PC receives the data packets returned from the control card, and analysis the received data packets to determine whether communication is successful.

##### **Serial setting:**

Baud rate: 115200, 57600, 38400, etc. selected by the selected tool.

Format string: "115200, 8, N, 1", you can change the baud rate value 115200 according to what you set to the controller.

##### **Packet data checksum**

Communication process using the packet data checksum to check the correctness of data transmission, Checksum calculations we should pay attention : Data checksum is cumulative each byte of data, use the 16bit (2 bytes) unsigned number to represent, so when the data validation is more than 0xFFFF, the checksum, and retain only 16bit value. For example,  $0xFFFFA + 0x09 = 0x0003$ .

#### ***Font size code***

| Font size code | Font size in pixels |
|----------------|---------------------|
| 0              | 8                   |
| 1              | 12                  |
| 2              | 16                  |
| 3              | 24                  |
| 4              | 32                  |
| 5              | 40                  |

|   |    |
|---|----|
| 6 | 48 |
| 7 | 56 |

### ***Font style code***

| Font style code | Font style describe |
|-----------------|---------------------|
| 0               | Default font style  |
| 1               | Font style 1        |
| 2               | Font style 2        |
| 3               | Font style 3        |
| 4               | Font style 4        |
| 5               | Font style 5        |
| 6               | Font style 6        |
| 7               | Font style 7        |

### ***Text color code***

#### **1-byte color value**

Max 8 colors. One bit for one basic color.

Bit 0: red color

Bit 1: green color

Bit 2: blue color

Other: not used

Example:

| Color value | Color  |
|-------------|--------|
| 1           | Red    |
| 2           | Green  |
| 3           | Yellow |
| 4           | Blue   |
| 7           | White  |

#### **3-byte color value**

RGB color, one byte for one basic color. It can express all kinds of color. Use each one byte to represent red、green、blue.

Byte 1: Red value of the color

Byte 2: Green value of the color

Byte 3: Blue value of the color

***Picture effect code***

| Code | Picture effect |
|------|----------------|
| 0    | Center         |
| 1    | Zoom           |
| 2    | Stretch        |
| 3    | tile           |

***Special effect for text and picture***

| Code | Effect                       |
|------|------------------------------|
| 0    | Draw                         |
| 1    | Open from left               |
| 2    | Open from right              |
| 3    | Open from center(Horizontal) |
| 4    | Open from center(Vertical)   |
| 5    | Shutter(vertical)            |
| 6    | Move to left                 |
| 7    | Move to right                |
| 8    | Move up                      |
| 9    | Move down                    |
| 10   | Scroll up                    |
| 11   | Scroll to left               |
| 12   | Scroll to right              |
| 13   | Flicker                      |
| 14   | Continuous scroll to left    |

|    |                                    |
|----|------------------------------------|
| 15 | Continuous scroll to right         |
| 16 | Shutter(horizontal)                |
| 17 | Clockwise open out                 |
| 18 | Anticlockwise open out             |
| 9  | Windmill                           |
| 20 | Windmill (anticlockwise)           |
| 21 | Rectangle forth                    |
| 22 | Rectangle entad                    |
| 23 | Quadrangle forth                   |
| 24 | Quadrangle endtad                  |
| 25 | Circle forth                       |
| 26 | Circle endtad                      |
| 27 | Open out from left up corner       |
| 28 | Open out from right up corner      |
| 29 | Open out from left bottom corner   |
| 30 | Open out from right bottom corner  |
| 31 | Bevel open out                     |
| 32 | AntiBevel open out                 |
| 33 | Enter into from left up corner     |
| 34 | Enter into from right up corner    |
| 35 | Enter into from left bottom corner |
| 36 | Enter into from lower right corner |

|    |                           |
|----|---------------------------|
| 37 | Bevel enter into          |
| 38 | AntiBevel enter into      |
| 39 | Horizontal zebra crossing |
| 40 | Vertical zebra crossing   |
| 41 | Mosaic(big)               |
| 42 | Mosaic(small)             |
| 43 | Radiation(up)             |
| 44 | Radiation(downwards)      |
| 45 | Amass                     |
| 46 | Drop                      |
| 47 | Combination(Horizontal)   |
| 48 | Combination(Vertical)     |
| 49 | Backout                   |
| 50 | Screwing in               |
| 51 | Chessboard(horizontal)    |
| 52 | Chessboard(vertical)      |
| 53 | Continuous scroll up      |
| 54 | Continuous scroll down    |
| 55 | Reserved                  |
| 56 | Reserved                  |
| 57 | Gradual bigger(up)        |
| 58 | Gradual smaller(down)     |
| 59 | Reserved                  |
| 60 | Gradual bigger(vertical)  |
| 61 | Flicker(horizontal)       |

|    |                             |
|----|-----------------------------|
| 62 | Flicker(vertical)           |
| 63 | Snow                        |
| 64 | Scroll down                 |
| 65 | Scroll from left to right   |
| 66 | Open out from top to bottom |
| 67 | Sector expand               |
| 68 | Reserved                    |
| 69 | Zebra crossing (horizontal) |
| 70 | Zebra crossing (Vertical)   |
|    |                             |
|    |                             |
|    |                             |
|    |                             |
|    |                             |
|    |                             |

When the random effect is expressed by one byte , the value is 255(0xFF) , when it is expressed by two bytes , the value is 32768(0x8000).

### ***Clock format and display content***

#### **Clock format :**

Represent by one byte :

bit 0: Signal timing(0: 12signal timing ; 1: 24 signal timing)

bit 1: Year by bit(0: 4 bit ; 1: 2 bit)

bit 2: Line folding(0: single-row ; 1: multi-row)

bit 3~5: Reserved(set to 0)

bit 6: Show time scale "Hour scale、 Minute scale"

bit 7: Reserved(set to 0)



**Clock display content :**

Represent by one byte :

Ascertain the display content by bit:

bit 7: pointer

bit 6: week

bit 5: second

bit 4: minute

bit 3: hour

bit 2: day

bit 1: month

bit 0: year

**Simple picture data format**
**Data composition :**

|           |                       |                         |                        |
|-----------|-----------------------|-------------------------|------------------------|
| Data head | Red<br>data(optional) | Green<br>data(optional) | Blue<br>data(optional) |
|-----------|-----------------------|-------------------------|------------------------|

**Data head description :**

|      |          |       |        |          |          |   |   |   |
|------|----------|-------|--------|----------|----------|---|---|---|
|      | 0        | 1     | 2      | 3        | 4        | 5 | 6 | 7 |
| 0x00 | identify | width | height | property | Reserved |   |   |   |

**Description :**

| Data name | Data size(byte) | Description                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identify  | 2               | Set to "I1".                                                                                                                                                                                                                                                                                                                                                            |
| Width     | 2               | The width of the picture, low byte previous(little endian)                                                                                                                                                                                                                                                                                                              |
| Height    | 2               | The height of the picture, low byte previous(little endian)                                                                                                                                                                                                                                                                                                             |
| Property  | 1               | The gray-scale and color of the picture<br>Bit0: Whether exist red data, exist when 1.<br>Bit1: Whether exist green data, exist when 1.<br>Bit2: Whether exist blue data, exist when 1.<br>Bit3: Reserved, set to 0.<br>Bit4~7: Gray-scale, only support 0 and 7 now.<br>0: 2 levels gray, Each latic data use 1 bit.<br>7: 256 levels gray, Each latic data use 8 bit. |

|          |   |                                                                                                                                    |
|----------|---|------------------------------------------------------------------------------------------------------------------------------------|
|          |   | Each line of the picture data is aligned by byte. As for 2 levels gray picture, when the line data is not enough for 8 bit, add 0. |
| Reserved | 1 | Set 0                                                                                                                              |

### Data description :

The color of the data is order by red、green、blue. If the identify bit of the property is 0, the color data is not exist.

For one color data, order by "left to right, top to bottom". First put the data to the first line, then second line and so on.

### Formatted text data format

#### Rich3 Text

Each character is 3 bytes, the specific meaning of each byte are as follows:

| Byte no | Byte data                                                                                                                                                                                                                                                              |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Said the color and font size: 4bits (1-7) represent the color (red green yellow blue purple cyan white), low 4bits (= 0 indicates that 8-point text; = 2 16-point text; = 3 24-point text; = 4 32point text; = 5 40-point text; = 6 48-point text; = 7 56-point text). |
| 2       | High byte of the text encoding. For single-byte characters, the value is 0.                                                                                                                                                                                            |
| 3       | Low byte of the text encoding. For single-byte characters, the value of its ASCII code.                                                                                                                                                                                |

## 2、Data packet format

### 2.1 RS232/RS485 data packet format :

#### 2.1.1 The data packet format of RS232/RS485 sending:

0xa5 0x68 0x32 ID 7B FF LL LH PO TP CC . . . . . SH SL 0xae

| Data        | Value | Length(Byte) | Description                        |
|-------------|-------|--------------|------------------------------------|
| Start code  | 0xa5  | 1            | The start of a packet              |
| Packet type | 0x68  | 1            | Recognition of this type of packet |



|                                          |                   |                 |                                                                                                                                                                                                                                                 |
|------------------------------------------|-------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Card type                                | 0x32              | 1               | Fixed Type Code                                                                                                                                                                                                                                 |
| Card ID                                  | 0x01~0xFE<br>0XFF | 1               | Control card ID, the screen No, valid values are as follows:<br>1 ~ 254: the specified card ID<br>0XFF: that group address, unconditionally receiving data                                                                                      |
| Protocol code                            | 0X7B              | 1               | Recognition of this type of potocol                                                                                                                                                                                                             |
| Additional information/confirmation mark | FF                | 1               | The meaning of bytes in the packet is sent, "Additional Information", is a packet plus instructions, and now only use the lowest:<br>bit 0: whether to return a confirmation, 1 to return and 0 not to return<br>bit1 ~ bi7: reserved, set to 0 |
| Packed data length<br>LL LH              | 0x0000~0xffff     | 2               | Two bytes, the length of the "CC ....." part content . Lower byte in the former                                                                                                                                                                 |
| Packet number<br>PO                      | 0x00~0x255        | 1               | When the packet sequence number is equal to when the last packet sequence number, indicating that this is the last one package.                                                                                                                 |
| Last packet number<br>TP                 | 0x00~0x255        | 1               | The total number of packages minus 1.                                                                                                                                                                                                           |
| Packet data                              | CC .....          | Variable-length | Command sub-code and data                                                                                                                                                                                                                       |
| Packet data checksum<br>SH SL            | 0x0000~0xffff     | 2               | Two bytes, checksum。 Lower byte in the former。 The sum of each byte from " Packet type " to “ Packet data” content                                                                                                                              |
| End code                                 | 0xae              | 1               | The end of a packet (Package tail)                                                                                                                                                                                                              |

### 2.1.2 The data packet format of the control card returned to RS232/RS485 sender:

0xa5 0x68 0x32 ID 7B FF LL LH PO TP CC . . . . . SH SL 0xae

| Data                        | Value             | Length(Byte)    | Description                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start code                  | 0xa5              | 1               | The start of a packet                                                                                                                                                                                                                                                                 |
| Packet type                 | <b>0xE8</b>       | 1               | Recognition of this type of packet<br>0xE8 = (0x68   0x80), for the app 3.2 or below return 0x68, app 3.3 or above return 0xe8 to same as other protocol( such as "set time" protocol), so you can ignore the highest bit (0x80), then it works for all app version                   |
| Card type                   | 0x32              | 1               | Fixed Type Code                                                                                                                                                                                                                                                                       |
| Card ID                     | 0x01~0xFE<br>0XFF | 1               | Control card ID, the screen No, valid values are as follows:<br>1 ~ 254: the specified card ID<br>0XFF: that group address, unconditionally receiving data                                                                                                                            |
| Protocol code               | 0X7B              | 1               | Recognition of this type of protocol.                                                                                                                                                                                                                                                 |
| Return value                | RR                | 1               | RR = 0x00: that successful;<br>RR = 0x01 ~ 0xFF: that the failure error code.<br>(0x01: checksum error)<br>(0x02: packet sequence error)<br>(Other: to be confirmed)<br>In addition, a certain period of time does not receive the returned data packet, said communication failures. |
| Packed data length<br>LL LH | 0x0000~0xffff     | 2               | Two bytes, the length of the "CC<br>. . . . . " part content . Lower byte in the former                                                                                                                                                                                               |
| Packet number<br>PO         | 0x00~0x255        | 1               | When the packet sequence number is equal to when the last packet sequence number, indicating that this is the last one package.                                                                                                                                                       |
| Last packet number<br>TP    | 0x00~0x255        | 1               | The total number of packages minus 1.                                                                                                                                                                                                                                                 |
| Packet data                 | CC<br>. . . . .   | Variable-length | Command sub-code and data                                                                                                                                                                                                                                                             |

|                                  |               |   |                                                                                                                      |
|----------------------------------|---------------|---|----------------------------------------------------------------------------------------------------------------------|
| Packet data<br>checksum<br>SH SL | 0x0000~0xffff | 2 | Two bytes, checksum. Lower byte in the former. The sum of each byte from " Packet type " to " Packet data" content . |
| End code                         | 0xae          | 1 | The end of a packet (Package tail)                                                                                   |

"Packet number", "Last packet number" in the return package was re-calculated by the number of packets returned.

### 2.1.3 RS232/RS485 packet data transcoding description:

The following process is done sending and receiving low-level functions, If you write your own PC side of the sending and receiving programs, you must implementation as below conventions. Use the without transcoding code to calculation checksum.

#### Send :

Between start code and end code, if there is 0xA5, 0xAA or 0xAE, it should be converted to two code :

0xa5 → 0xaa 0x05. The purpose is to avoid the same with the start character 0xa5

0xae → 0xaa 0x0e. The purpose is to avoid the same with the end of the symbol 0xae.

0xaa → 0xaa 0x0a. The purpose is to avoid the same with the escape character 0xaa.

#### Receive :

Received symbol 0xa5, said that the beginning of a packet

Received symbol 0xae, said that the end of a packet

When PC receive data from controller, if there is 0xA5, 0xAA or 0xAE, it should convert two code to one code, specifically for

0xaa 0x05 → 0xa5

0xaa 0x0e → 0xae

0xaa 0x0a → 0xaa

## 2.2 Network data packet format

### 2.2.1 The data packet format of network sending

| Data                                     | Value                   | Length(Byte) | Description                                                                                                                                                                                                                                     |
|------------------------------------------|-------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID Code                                  | 0x00000000 ~ 0xffffffff | 4            | Control card ID, high byte in the former.<br>Need to set to the same value on the card.                                                                                                                                                         |
| Network data length                      | 0x0000 ~ 0xffff         | 2            | The byte length that from "Packet type" to "Packet data checksum".                                                                                                                                                                              |
| Reservation                              | 0x0000                  | 2            | Reservations. Fill in 0                                                                                                                                                                                                                         |
| Packet type                              | 0x68                    | 1            | Recognition of this type of packet                                                                                                                                                                                                              |
| Card type                                | 0x32                    | 1            | Fixed Type Code                                                                                                                                                                                                                                 |
| Card ID                                  | 0x01~0xFE<br>0XFF       | 1            | Control card ID, the screen No, valid values are as follows:<br>1 ~ 254: the specified card ID<br>0XFF: that group address, unconditionally receiving data                                                                                      |
| Protocol code                            | 0X7B                    | 1            | Recognition of this type of protocol.                                                                                                                                                                                                           |
| Additional information/confirmation mark | FF                      | 1            | The meaning of bytes in the packet is sent, "Additional Information", is a packet plus instructions, and now only use the lowest:<br>bit 0: whether to return a confirmation, 1 to return and 0 not to return<br>bit1 ~ bi7: reserved, set to 0 |
| Packed data length<br>LL LH              | 0x0000~0xffff           | 2            | Two bytes, the length of the "CC . . . . . " part content . Lower byte in the former                                                                                                                                                            |
| Packet number<br>PO                      | 0x00~0x255              | 1            | When the packet sequence number is equal to when the last packet sequence number, indicating that this is the last one package.                                                                                                                 |
| Last packet number                       | 0x00~0x255              | 1            | The total number of packages minus 1.                                                                                                                                                                                                           |

|                               |                   |                 |                                                                                                                       |
|-------------------------------|-------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| TP                            |                   |                 |                                                                                                                       |
| Packet data                   | CC<br>o o o o o o | Variable-length | Command sub-code and data                                                                                             |
| Packet data checksum<br>SH SL | 0x0000~0xffff     | 2               | Two bytes, checksum. Lower byte in the former. The sum of each byte from " Packet type " to " Packet data " content . |

The network packet data does not need to do transcoding processing.

### 2.2.2 The data packet format of the control card returned to network sender

| Data                | Value                   | Length(Byte) | Description                                                                                                                                                                                                                                                         |
|---------------------|-------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID Code             | 0x00000000 ~ 0xffffffff | 4            | Control card ID, high byte in the former.<br>Need to set to the same value on the card.                                                                                                                                                                             |
| Network data length | 0x0000 ~ 0xffff         | 2            | The byte length that from "Packet type" to "Packet data checksum".                                                                                                                                                                                                  |
| Reservation         | 0x0000                  | 2            | Reservations. Fill in 0                                                                                                                                                                                                                                             |
| Packet type         | 0xE8                    | 1            | Recognition of this type of packet<br>0xE8 = (0x68   0x80), for the app 3.2 or below return 0x68, app 3.3 or above return 0xe8 to same as other protocol( such as "set time" protocol), so you can ignore the highest bit (0x80), then it works for all app version |
| Card type           | 0x32                    | 1            | Fixed Type Code                                                                                                                                                                                                                                                     |
| Card ID             | 0x01~0xFE<br>0XFF       | 1            | Control card ID, the screen No, valid values are as follows:<br>1 ~ 254: the specified card ID<br>0XFF: that group address, unconditionally receiving data                                                                                                          |
| Protocol code       | 0X7B                    | 1            | Recognition of this type of protocol.                                                                                                                                                                                                                               |
| Return value        | RR                      | 1            | RR = 0x00: that successful;<br>RR = 0x01 ~ 0xFF: that the failure error code.<br>(0x01: checksum error)                                                                                                                                                             |



|                               |                 |                 |                                                                                                                                                                            |
|-------------------------------|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               |                 |                 | (0x02: packet sequence error)<br>(Other: to be confirmed)<br>In addition, a certain period of time does not receive the returned data packet, said communication failures. |
| Packed data length<br>LL LH   | 0x0000~0xffff   | 2               | Two bytes, the length of the "CC . . . . . " part content . Lower byte in the former                                                                                       |
| Packet number<br>PO           | 0x00~0x255      | 1               | When the packet sequence number is equal to when the last packet sequence number, indicating that this is the last one package.                                            |
| Last packet number<br>TP      | 0x00~0x255      | 1               | The total number of packages minus 1.                                                                                                                                      |
| Packet data                   | CC<br>. . . . . | Variable-length | Command sub-code and data                                                                                                                                                  |
| Packet data checksum<br>SH SL | 0x0000~0xffff   | 2               | Two bytes, checksum. Lower byte in the former. The sum of each byte from " Packet type " to " Packet data " content .                                                      |

"Packet number", "Last packet number" in the return package was re-calculated by the number of packets returned.

The network packet data does not need to do transcoding processing.

### 2.3 Command sub-code and data : CC. . . . .

CC : A sub-byte instruction code, specifying the meaning of the data.

. . . . . : Data content for different sub-code instructions, there are different elements.

If the data needs to be divided into several packages, command sub-code only in the first data packet appears, the other only contains the data content of data packets.



### 2.2.1 Command sub-code includes:

#### General protocol command

| Command code(CC) | sub- | meanings                                                      |
|------------------|------|---------------------------------------------------------------|
| 0x01             |      | Division of display window (area)                             |
| 0x02             |      | To send text data to a specified window                       |
| 0x03             |      | To send image data to the specified window                    |
| 0x04             |      | Static text data sent to the specified window                 |
| 0x05             |      | To send clock data to the specified window                    |
| 0x06             |      | Exit show to return to play within the program                |
| 0x07             |      | Save / clear the data                                         |
| 0x08             |      | Select play stored program (single-byte)                      |
| 0x09             |      | Select play t stored program (double-byte)                    |
| 0x0a             |      | Set variable value                                            |
| 0x0b             |      | Select play single stored program, and set the variable value |
| 0x0c             |      | Set global display area                                       |
| 0x0d             |      | Push user variable data                                       |
| 0x0e             |      | Set timer control                                             |
| 0x0f             |      | Set the global display area and variable values               |
| 0x12             |      | Send pure text to the specified window                        |

#### Program template command

| Command code(CC) | sub- | meanings                           |
|------------------|------|------------------------------------|
| 0x81             |      | Set program template command       |
| 0x82             |      | In or out program template command |
| 0x83             |      | Query program template command     |
| 0x84             |      | Delete program command             |
| 0x85             |      | Send text to special window        |

|      |                                                                              |
|------|------------------------------------------------------------------------------|
| 0x86 | Send picture to special window                                               |
| 0x87 | Clock / temperature display in the specified window of the specified program |
| 0x88 | Send alone program                                                           |
| 0x89 |                                                                              |
| 0x8a | Set program property                                                         |
| 0x8b | Set play plan                                                                |
| 0x8c | Delete play plan                                                             |
| 0x8d | Query play plan                                                              |
|      |                                                                              |

### 2.2.2 The specific definition of command sub-code and data:

#### *Division of display window : CC=0x01:*

| Data Items                          | Value         | Length(byte) | Description                                                        |
|-------------------------------------|---------------|--------------|--------------------------------------------------------------------|
| CC                                  | 0x01          | 1            | Note This command is divided into display window (area)            |
| Window Number                       | 0x01~0x08     | 1            | The window should be divided into the number of valid value 1 ~ 8. |
| Window 1 X-coordinate<br>XH XL      | 0x0000~0xffff | 2            | Window x-coordinate, high byte in the former                       |
| Window 1 Y-coordinate<br>YH YL      | 0x0000~0xffff | 2            | Window y-coordinate, high byte in the former                       |
| The width of the window<br>1 WH WL  | 0x0000~0xffff | 2            | The width of the window, high byte in the former                   |
| The height of the window<br>1 HH HL | 0x0000~0xffff | 2            | The height of the window, high byte in the former                  |
| .....                               |               |              |                                                                    |
| Window N X-coordinate<br>XH XL      | 0x0000~0xffff | 2            | Window x-coordinate, high byte in the former                       |

|                                     |               |   |                                                   |
|-------------------------------------|---------------|---|---------------------------------------------------|
| Window N<br>Y-coordinate<br>YH YL   | 0x0000~0xffff | 2 | Window y-coordinate, high byte in the former      |
| The width of the window<br>N WH WL  | 0x0000~0xffff | 2 | The width of the window, high byte in the former  |
| The height of the window<br>N HH HL | 0x0000~0xffff | 2 | The height of the window, high byte in the former |

- Based on the above definition, requires 8 bytes for each window's location and size, then divided into N windows, data on a total of  $2 + 8 * N$  bytes

**Send text data to a specified window : CC=0x02:**

| Data Items | Value         | Length(byte)    | Description                                                                                |
|------------|---------------|-----------------|--------------------------------------------------------------------------------------------|
| CC         | 0x02          | 1               | Description This is a text data packet                                                     |
| Window No  | 0x00~0x07     | 1               | The window sequence number, valid values 0 ~ 7.                                            |
| Mode       | 1             | 1               | Refer to Special effect for text and picture                                               |
| Alignment  | 0~2           | 1               | 0: Left-aligned<br>1: Horizontal center<br>2: Right-aligned                                |
| Speed      | 1 ~ 100       | 1               | The smaller the value, the faster                                                          |
| Stay time  | 0x0000~0xffff | 2               | High byte in the former. Unit: second.                                                     |
| String     |               | Variable-length | Every 3 bytes to represent a character. Refer to Rich3 text of Formatted text data format. |

**Send image data to a specified window : CC=0x03:**

| Data Items | Value         | Length(byte) | Description                                                               |
|------------|---------------|--------------|---------------------------------------------------------------------------|
| CC         | 0x03          | 1            | Description This is a image data packet                                   |
| Window No  | 0x00~0x07     | 1            | The window sequence number, valid values 0 ~ 7.                           |
| Mode       | 0x00          | 1            | 0x00: Draw                                                                |
| Speed      | 1             | 1            | The smaller the value, the faster. Now appears that this value is invalid |
| Stay time  | 0x0000~0xffff | 2            | High in the former. Units of seconds.                                     |

|                          |               |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|---------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Image Data Format        | 0x01          | 1               | 0x01: gif image file format<br>0x02: gif image file references.<br>0x03: picture package picture reference.<br>0x04: simple image format.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Image Display X Position | 0x0000~0xffff | 2               | Began to show the location of X coordinate. Relative upper-left corner the window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Image Display Y Position | 0x0000~0xffff | 2               | Began to show the location of Y coordinate. Relative upper-left corner the window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Image Data               |               | Variable-length | <p>According to "image data format" is defined to determine the meaning of the data.</p> <p>Image data format is 0x01: gif image file of the actual data, which contains the image width, height and other information;</p> <p>Image data format is 0x02: the gif image file name stored in the control card.</p> <p>Image data format is 0x03: The image package file name and image number that stored in the controller. The middle separated by spaces. For example, "images.rpk 1"</p> <p>Image data format is 0x04 : Simple picture data, see the description format.</p> |

**Send static text : CC=0x04:**

| Data Items | Value     | Length(byte) | Description                                   |
|------------|-----------|--------------|-----------------------------------------------|
| CC         | 0x04      | 1            | Description of the data packet is static text |
| Window NO  | 0x00~0x07 | 1            | Window sequence number, valid values 0 to 7   |
| Data type  | 1         | 1            | 0x01: Simple text data                        |

|                        |               |                 |                                                                                                     |
|------------------------|---------------|-----------------|-----------------------------------------------------------------------------------------------------|
| The level of alignment | 0~2           | 1               | 0: left Alignment<br>1: center Alignment<br>2: right Alignment                                      |
| Display area X         | 0x0000~0xffff | 2               | The X coordinate of upper left corner of the display area. Upper left corner of the window relative |
| Display area Y         | 0x0000~0xffff | 2               | The Y coordinate of upper left corner of the display area. Upper left corner of the window relative |
| Display area width     | 0x0000~0xffff | 2               | The width of display area. High byte in the former.                                                 |
| Display area height    | 0x0000~0xffff | 2               | The height of display area. High byte in the former.                                                |
| Font                   |               | 1               | Bit0~3: font size<br>Bit4~6: font style<br>Bit7: Reserved                                           |
| Text color R           | 0~255         | 1               | The red color component                                                                             |
| Text color G           | 0~255         | 1               | The green color component                                                                           |
| Text color B           | 0~255         | 1               | The blue color component                                                                            |
| Text                   |               | Variable-length | Text string to the end of 0x00.                                                                     |

**Send clock : CC=0x05:**

| Data Items | Value     | Length(byte) | Description                                                                                                                           |
|------------|-----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| CC         | 0x05      | 1            | Description of the data packet is clock                                                                                               |
| Window NO  | 0x00~0x07 | 1            | Window sequence number, valid values 0 to 7                                                                                           |
| Stay time  |           | 2            | Stay time in second. High byte in the former                                                                                          |
| Calendar   |           | 1            | 0: Gregorian calendar date and time<br>1: Lunar date and time<br>2: Chinese lunar solar terms<br>3: Lunar time and date + Solar Terms |
| Format     |           | 1            | Format: Format<br>bit 0: when the system (0: 12 hour; 1: 24                                                                           |



|              |       |                 |                                                                                                                                                                              |
|--------------|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |       |                 | hours system)<br>bit 1: Year digit (0: 4; 1: 2)<br>bit 2: Branch (0: single; 1: multi-line)<br>bit 3 ~ 7: reserved (set to 0)                                                |
| Content      |       | 1               | By bit to determine the content to display.<br>bit 7: Pointer<br>bit 6: weeks<br>bit 5: seconds<br>bit 4: minute<br>bit 3: hour<br>bit 2: day<br>bit 1: month<br>bit 0: year |
| Font         |       | 1               | Bit0~3: font size                                                                                                                                                            |
| Text color R | 0~255 | 1               | The red color component                                                                                                                                                      |
| Text color G | 0~255 | 1               | The green color component                                                                                                                                                    |
| Text color B | 0~255 | 1               | The blue color component                                                                                                                                                     |
| Text         |       | Variable-length | Text string to the end of 0x00.                                                                                                                                              |

**Exit show and return to play within the program : CC=0x06:**

| Data Items | Value | Length(byte) | Description                      |
|------------|-------|--------------|----------------------------------|
| CC         | 0x06  | 1            | Play programs stored on the card |

**Save/clear the data : CC=0x07:**

| Data Items | Value     | Length(byte) | Description                                                          |
|------------|-----------|--------------|----------------------------------------------------------------------|
| CC         | 0x07      | 1            | The data packet is a request control card to save data in the window |
| Save/clear | 0x00/0x01 | 1            | 0x00: save data to flash.<br>0x01: Clear flash data                  |
| Reserve    | 0x00 0x00 | 2            | Reserved for later expansion                                         |

**Select play stored program (single-byte) : CC=0x08**

| Data Items | Value | Length(byte) | Description                       |
|------------|-------|--------------|-----------------------------------|
| CC         | 0x08  | 1            | Description of the data packet is |

|                          |            |                 |                                                                                                                    |
|--------------------------|------------|-----------------|--------------------------------------------------------------------------------------------------------------------|
|                          |            |                 | stored program data select play(single-byte)                                                                       |
| Options                  |            | 1               | Bit0: Whether to save select play message to flash. 0 not to save, 1 save.<br>Bit1~7: Reserved, set to 0           |
| The number of programs   | 1~255 or 0 | 1               | The program number that to be selected to play, if the number is 0,the controller will exit the select play state. |
| The program number table | 1~255      | Variable-length | The program number list,1 byte for each program. Exceed the number of programs stored program number is ignored    |

**Select play stored program (double-byte) : CC=0x09**

| Data Items               | Value      | Length(byte)    | Description                                                                                                                                                       |
|--------------------------|------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC                       | 0x09       | 1               | Description of the data packet is stored program data select play(double-byte)                                                                                    |
| Options                  |            | 1               | Bit0: Whether to save select play message to flash. 0 not to save, 1 save.<br>Bit1~7: Reserved, set to 0                                                          |
| The number of programs   | 1~512 or 0 | 2               | The program number that to be selected to play, the max value is 512, high byte in the former. if the number is 0,the controller will exit the select play state. |
| The program number table | 1~65535    | Variable-length | The program number list,2 bytes for each program. Exceed the number of programs stored program number is ignored                                                  |

**Set variable value : CC=0x0a**

| Data Items | Value | Length(byte) | Description                                                   |
|------------|-------|--------------|---------------------------------------------------------------|
| CC         | 0x0a  | 1            | Description of the data packet is the data set variable value |
| Options    |       | 1            | Bit0: Whether to save all variable value                      |

|                                               |           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------|-----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               |           |                 | to flash, 0 not to save, 1 save.<br>Bit1: Whether to clear all variable value before save, 0 not to clear, 1 clear.<br>Bit1~7: Reserved, set to 0                                                                                                                                                                                                                                                                                                                                                                                                             |
| Variable number and allow cross-variable zone | 1~100     | 1               | Bit0~6 : The variable number<br>Bit7 : Whether to allow cross-variable zone setting. 0 is not permitted; 1 is permit<br>Corresponds to a variable number of each variable area size of each variable region is 32 bytes. Multiple continuous variables can be linked to a variable area used, occupied area of the variable number of variables can not be used.<br>When does not allow cross-variable area, more than 32 bytes of data are discarded ; When allow cross-variable area, calculate the length of the data area to use the number of variables. |
| Variable data length table                    | n (0~255) | Variable-length | Specified by the order of bytes of data for each variable. The length of variable number and data is (1 + n) bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Variable number and data                      |           | Variable-length | The first byte is a variable number, followed by a specified length of variable data                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Note :**

Valid values for the variable number is 1~100. Number of variables corresponding to each variable area can store 32 bytes of data, a number of continuous variable area can be used together for a variable, the variable area occupied number of variables can not be used.

When variable values are not updated and just save the variable value to the FLASH, it can set the " Variable number " of the value of 0, set the " Options " to save

**Select play single stored program and set the variable value : CC=0x0b**

| Data Items                                    | Value     | Length(byte)    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|-----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC                                            | 0x0b      | 1               | Description of the data packet is the data select single program and set variable value                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Options                                       |           | 1               | Bit0: Whether to save program number to flash<br>0 not to save, 1 save.<br>Bit1: Whether to save all variable value to flash , 0 not to save, 1 save.<br>Bit2: Whether to clear all variable value before save, 0 not to clear, 1 clear.<br>Bit3~7: Reserved, set to 0                                                                                                                                                                                                                                                                                        |
| Program numbers                               | 1~65535   | Variable-length | The program number list, 2 bytes for each program. Exceed the number of programs stored program number is ignored                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Variable number and allow cross-variable zone | 1~100     | 1               | Bit0~6 : The variable number<br>Bit7 : Whether to allow cross-variable zone setting. 0 is not permitted; 1 is permit<br>Corresponds to a variable number of each variable area size of each variable region is 32 bytes. Multiple continuous variables can be linked to a variable area used, occupied area of the variable number of variables can not be used.<br>When does not allow cross-variable area, more than 32 bytes of data are discarded ; When allow cross-variable area, calculate the length of the data area to use the number of variables. |
| Variable data length table                    | n (0~255) | Variable-length | Specified by the order of bytes of data for each variable. The length of variable number and data is (1 + n) bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Variable                                      |           | Variable-       | The first byte is a variable number,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                 |  |        |                                                 |
|-----------------|--|--------|-------------------------------------------------|
| number and data |  | length | followed by a specified length of variable data |
|-----------------|--|--------|-------------------------------------------------|

**Note :**

Valid values for the variable number is 1~100. Number of variables corresponding to each variable area can store 32 bytes of data, a number of continuous variable area can be used together for a variable, the variable area occupied number of variables can not be used.

When variable values are not updated and just save the variable value to the FLASH, it can set the " Variable number " of the value of 0, set the " Options " to save.

**Set global display area : CC=0x0c**

| Data Items      | Value | Length(byte)  | Description                                                                                            |
|-----------------|-------|---------------|--------------------------------------------------------------------------------------------------------|
| CC              | 0x0c  | 1             | Describe the packet is the data which to set the global display zone.                                  |
| Options         |       | 1             | Bit0: Whether to save the setup to FLASH<br>0 not to save, 1 save.<br>Bit1~7: Reserved, set value to 0 |
| ZoneArea count  | 1~8   | 1             | The count of global display zone which is to be set.<br>Cancel all the zone when zone count is 0       |
| Synchronization |       | 1             | Bit0: Synchronous display. 0 not synchronous, 1 synchronous.<br>Bit1~7: Reserved                       |
| Retention       |       | 2             | Set value 0                                                                                            |
| Zone Definition |       | Zone Count*16 | The specific definition of global display zone                                                         |

**Area Definition :** (16 bytes each item)

The first byte is zone type, available zone types:

| Vale | Type                                       |
|------|--------------------------------------------|
| 1    | Display the variable's specify text        |
| 2    | Display the variable's specify file (.gif) |
| 6    | Display hint text of other zone            |
| 7    | Display stopwatch timer value              |



|       |          |
|-------|----------|
| other | Reserved |
|-------|----------|

**Type = 1**

|      |      |          |   |   |   |   |    |   |    |   |       |     |      |   |      |       |
|------|------|----------|---|---|---|---|----|---|----|---|-------|-----|------|---|------|-------|
|      | 0    | 1        | 2 | 3 | 4 | 5 | 6  | 7 | 8  | 9 | A     | B   | C    | D | E    | F     |
| 0x00 | Type | reserved | x |   | y |   | cx |   | cy |   | start | end | stay |   | font | align |

Explanation :

| Data Name | Data Size (byte) | Description                                                                                                                                                                                     |
|-----------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | 1                | 1 : Display the variable's specify text                                                                                                                                                         |
| x         | 2                | Zone start point X. High byte previous                                                                                                                                                          |
| y         | 2                | Zone start point Y. High byte previous                                                                                                                                                          |
| cx        | 2                | Zone width. High byte previous                                                                                                                                                                  |
| cy        | 2                | Zone Height. High byte previous                                                                                                                                                                 |
| Start     | 1                | Start variable number, valid value 1~100                                                                                                                                                        |
| End       | 1                | End variable number, valid value 1~100                                                                                                                                                          |
| stay      | 2                | The stay time when display each valid variable's content, the unit is second.<br>High byte previous                                                                                             |
| font      | 1                | Font size and color<br>Bit0~2: font size(8,12,16,24,32,40,48,56)<br>Bit3: Color invert<br>Bit4: Red value of color<br>Bit5: Green value of color<br>Bit6: Blue value of color<br>Bit7: Reserved |
| align     | 1                | Text alignment                                                                                                                                                                                  |

All "reserved" values need to be set to 0

**Type = 2**

|      |      |          |   |   |   |   |    |   |    |   |       |     |      |      |   |          |
|------|------|----------|---|---|---|---|----|---|----|---|-------|-----|------|------|---|----------|
|      | 0    | 1        | 2 | 3 | 4 | 5 | 6  | 7 | 8  | 9 | A     | B   | C    | D    | E | F        |
| 0x00 | Type | reserved | x |   | y |   | cx |   | cy |   | start | end | stay | mode |   | reserved |

Explanation :

| Data Name | Data Size (byte) | Description                                   |
|-----------|------------------|-----------------------------------------------|
| Type      | 1                | 2: Display the variable's specify file (.gif) |
| x         | 2                | Zone start point X. High byte previous        |
| y         | 2                | Zone start point Y. High byte previous        |
| cx        | 2                | Zone width. High byte previous                |

|          |   |                                                                                                     |
|----------|---|-----------------------------------------------------------------------------------------------------|
| cy       | 2 | Zone Height. High byte previous                                                                     |
| Start    | 1 | Start variable number, valid value 1 ~ 100                                                          |
| End      | 1 | End variable number, valid value 1 ~ 100                                                            |
| stay     | 2 | The stay time when display each valid variable's content, the unit is second.<br>High byte previous |
| mode     | 1 | Image draw mode:<br>0: left top                                                                     |
| Reserved | 1 |                                                                                                     |

All "reserved" values need to be set to 0

**Type = 7**

|      |      |          |   |   |   |   |    |   |    |   |      |        |          |   |   |   |
|------|------|----------|---|---|---|---|----|---|----|---|------|--------|----------|---|---|---|
|      | 0    | 1        | 2 | 3 | 4 | 5 | 6  | 7 | 8  | 9 | A    | B      | C        | D | E | F |
| 0x00 | Type | reserved | x |   | y |   | cx |   | cy |   | font | format | reserved |   |   |   |

Explanation :

| Data Name | Data Size (byte) | Description                                                                                                                                                                                     |
|-----------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | 1                | 7: Display stopwatch timer value                                                                                                                                                                |
| x         | 2                | Zone start point X. High byte previous                                                                                                                                                          |
| y         | 2                | Zone start point Y. High byte previous                                                                                                                                                          |
| cx        | 2                | Zone width. High byte previous                                                                                                                                                                  |
| cy        | 2                | Zone Height. High byte previous                                                                                                                                                                 |
| font      | 1                | Font size and color<br>Bit0~2: font size(8,12,16,24,32,40,48,56)<br>Bit3: Color invert<br>Bit4: Red value of color<br>Bit5: Green value of color<br>Bit6: Blue value of color<br>Bit7: Reserved |
| format    | 1                | 0: "mm:ss"<br>1: "mm:ss:nn"                                                                                                                                                                     |
| Reserved  | 4                |                                                                                                                                                                                                 |

All "reserved" values need to be set to 0

**Push and set the variable value : CC=0x0d**

| Data Item                | Value | Lenght(byte) | Description                                                                                                                                                                                                           |
|--------------------------|-------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC                       | 0x0d  | 1            | Describe the package is the data which push and set the variable value                                                                                                                                                |
| Options                  |       | 1            | Bit0: Whether to save all the variable value to FLASH. 0 not save, 1 save.<br>Bit1: direction. 0 push back, 1 push forward<br>Bit 2~3: retention, set to 0<br>Bit4~7: push count. +1 means the variable count of push |
| Variable area count      |       | 1            | Bit0~6: The variable count for push 1~100<br>Bit7: Retention, set 0                                                                                                                                                   |
| Variable data length     |       | 1            | Specify variable data's byte.<br>The total length of variable number and data is (n+1) byte                                                                                                                           |
| Variable number and data |       | changeable   | The first byte is variable number, follow by specify length variable data.                                                                                                                                            |

**Set Timer : CC=0x0e**

| Data Item    | Value | Lenght(byte) | Description                                                                                                                                                                               |
|--------------|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC           | 0x0e  | 1            | Describe the package is the data of set stopwatch                                                                                                                                         |
| Select Timer |       | 1            | Select the Timer by bite. Bit value 1 means the Timer valid<br>Bit0: Timer 1.<br>Bit1: Timer 2.<br>Bit3: Timer 3.<br>Bit4: Timer 4.<br>Bit5: Timer 5.<br>Bit6: Timer 6.<br>Bit7: Timer 7. |
| Action       |       | 1            | 1 : Initialize Timer<br>2 : Reset Timer<br>3 : Startup Timer                                                                                                                              |

|          |  |   |                                                                                                   |
|----------|--|---|---------------------------------------------------------------------------------------------------|
|          |  |   | 4 : Pause Timer<br>5 : Save the setup of Timer<br>Other : Retention                               |
| property |  | 1 | Have different value according to the Action.<br>Check the detail information in the below table. |
| Value    |  | 4 | Have different value according to the Action.<br>Check the detail information in the below table. |

**The description of all Actions and the correspondence property and value**

| Action                  | Description | Property                                                                                                   | Value                                                                                                                                                                                              |
|-------------------------|-------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initialize Timer        |             | Bit0: 0 Time, 1 Countdown<br>Bit1: 0 Pause, 1 start immediately<br>Bit2~3: retention<br>Bit4~7: Time count | High byte previous.<br>The initialization value of countdown, measure time by millisecond.<br>The value retention when time, set to 0                                                              |
| Reset Timer             |             | Bit0: 0 Use old value, 1 Use new value<br>Bit1: 0 Pause, 1 start immediately<br>Bit2~3: Retention          | High byte previous.<br>Countdown: Use as a new initialization value when the property is set to use new value.<br>Ignore when the property is set to use the old value.<br>Time: Retention, set 0. |
| Start Timer             |             | Reserved, set 0                                                                                            | Retention, set 0                                                                                                                                                                                   |
| Pause Timer             |             | Retention, set 0                                                                                           | Retention, set 0                                                                                                                                                                                   |
| Save the setup of Timer |             | Retention, set 0                                                                                           | Retention, set 0                                                                                                                                                                                   |

**Set the global display area and variable values: CC=0x0f**

By the command, set the global display area and variable values

| Data Item                                     | Value | Lenght(byte)     | Description                                                                                                                                                                                                |
|-----------------------------------------------|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC                                            | 0x0f  | 1                | Describe the package is the data which to set the global display area and variable values                                                                                                                  |
| Effective control                             |       | 2                | Play times: High byte first.<br>The value of 0 has been effective.<br>Bit15: reserved, fill in 0<br>Bit0 to 14: play times                                                                                 |
| Reserved                                      |       | 2                | Reserved, fill in 0                                                                                                                                                                                        |
| Area option                                   |       | 1                | Bit0: Whether to save the setting to the flash , 0 : Don't save, 1 : Save.<br>Bit1~3: Reserved, fill in 0<br>Bit4: Whether to clear the others defined global display area<br>Bit5 ~7: Reserved, fill in 0 |
| Area number                                   | 1 ~8  | 1                | To set the number of global zone                                                                                                                                                                           |
| Area no                                       | 1 ~8  | Area number      | Bit0~3: Specified in the global area no. Valid values are 1 to 8. If the current no have used, then overwrite the original area information.<br>Bit 4~7: Reserved, fill in 0                               |
| Area definition                               |       | 16 x Area number | Specific definition of the global display area. Specific definition see CC = 0x0c (that set the global display area).                                                                                      |
| Variable Options                              |       | 1                | Bit0: Whether to save all variable value to flash, 0 not to save, 1 save.<br>Bit1: Whether to clear all variable value before save, 0 not to clear, 1 clear.<br>Bit1~7: Reserved, set to 0                 |
| Variable number and allow cross-variable zone |       | 1                | Bit0~6 : The variable number<br>Bit7 : Whether to allow cross-variable zone setting. 0 is not permitted; 1 is permit<br>Corresponds to a variable number of each variable area size of each variable       |



|                            |           |                 |                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|-----------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |           |                 | region is 32 bytes. Multiple continuous variables can be linked to a variable area used, occupied area of the variable number of variables can not be used.<br>When does not allow cross-variable area, more than 32 bytes of data are discarded ; When allow cross-variable area, calculate the length of the data area to use the number of variables. |
| Variable data length table | n (0~255) | Variable-length | Specified by the order of bytes of data for each variable. The length of variable number and data is (1 + n) bytes.                                                                                                                                                                                                                                      |
| Variable number and data   |           | Variable-length | The first byte is a variable number, followed by a specified length of variable data                                                                                                                                                                                                                                                                     |

Note : After use this command, The global area automatically becomes synchronized display.

### **Send pure text to specified window : CC=0x12**

| Data Item | Value         | Lenght(byte) | Description                                                                                                                                                                                                                                                   |
|-----------|---------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC        | 0x12          | 1            | Send pure text to the specified window                                                                                                                                                                                                                        |
| Window no | 0x00~0x07     | 1            | The window sequence number, valid values 0 ~ 7.                                                                                                                                                                                                               |
| Effect    | 1             | 1            | See: Special effect for text and picture                                                                                                                                                                                                                      |
| Alignment | 0~2           | 1            | Bit0~1: the horizontal alignment (<br>0: Left-aligned<br>1: Horizontal center<br>2: Right- aligned)<br>Bit2~3: vertical alignment (<br>0:Top-aligned<br>1:Vertically center,<br>2: Bottom- aligned)<br>Bit4~6: Reserved, set to 0<br>Bit7: Reserved, set to 0 |
| Speed     | 1~100         | 1            | The smaller the value, the faster                                                                                                                                                                                                                             |
| Stay time | 0x0000~0xffff | 2            | High byte in the former. Units of                                                                                                                                                                                                                             |

|              |       |                 |                                                                                                                 |
|--------------|-------|-----------------|-----------------------------------------------------------------------------------------------------------------|
|              |       |                 | <a href="#">seconds.</a>                                                                                        |
| Font         |       | 1               | Bit0~3: font size,see : Font size code<br>Bit4~6 :font style,see :Font style code<br>Bit7 : reserved (set to 0) |
| Text color R | 0~255 | 1               | The red color component                                                                                         |
| Text color G | 0~255 | 1               | The green color component                                                                                       |
| Text color B | 0~255 | 1               | The blue color component                                                                                        |
| Text         |       | Variable-length | Text string to the end of 0x00.                                                                                 |

Example 1 : A5 68 32 01 7B 01 0f 00 00 00 12 00 00 00 00 00 03 02 ff 00 00 61 62 63 00 62 03 AE

Example 2 : A5 68 32 01 7B 01 13 00 00 00 12 00 00 01 00 00 03 02 ff 00 00 61 62 63 0d 63 62 61 00 9a 04 AE

### 2.2.3 Detail of program template command code and data

The program template agreement is a set of relatively independent of the agreement, the basic concept of this agreement are as follows:

"Program" is a standalone player within a certain time, a message is displayed on the screen can be divided into multiple regions, each region can be specified individually.

The maximum number of programs is 100, corresponding to an effective program number from 1 to 100. Send program information, new information covering the same program number.

#### **Set program template : CC=0x81:**

| Data Item      | Value | Lenght(byte) | Description                                                                                                                                           |
|----------------|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC             | 0x81  | 1            | Describe the package is the data which to set program template.                                                                                       |
| Color and gray |       | 1            | Bit0: Red mark<br>Bit1: Green mark<br>Bit2: Blue mark<br>Bit3: Reserved<br>Bit4~6: Gray level<br>0: 2 level gray, 7: 256 level gray<br>Bit7: Reserved |
| Screen width   |       | 2            | High byte first.                                                                                                                                      |
| Screen height  |       | 2            | High byte first.                                                                                                                                      |
| Window number  |       | 1            | The display window number, the maximum number is 10                                                                                                   |
| Options        |       | 1            | Bit0: Forced into the program template                                                                                                                |

|                   |                         |           |                                                                                                                                                                                                                                                                                              |
|-------------------|-------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                         |           | run<br>Bit1: Save the template position. 0: user disk, 1: system disk. If the template is saved to the system tray, the original template of the user tray is cleared; if the template is saved to the user's disk, the original template of the system disk is cleared.<br>Bit2~7: Reserved |
| Default parameter | Stay time/ Scroll times | 2         | Stay time/Scroll times: High byte first. When the show effect is scroll , it means scroll times(0 scroll one times,1 scroll two times,...) , for others , it means stay time, unit is second.                                                                                                |
|                   | Speed                   | 1         | The smaller the faster                                                                                                                                                                                                                                                                       |
|                   | Font size               | 1         | Bit 0~3:Font size, Font size code<br>Bit 4~6:Font style , Font style code                                                                                                                                                                                                                    |
|                   | Font color.             | 1         | Font color. Text color code                                                                                                                                                                                                                                                                  |
|                   | Show effect             | 1         | Show effect. Special effect for text and picture                                                                                                                                                                                                                                             |
|                   | Picture mode            | 1         | Picture mode. Picture effect code                                                                                                                                                                                                                                                            |
|                   | Clock Format            | 1         | Clock Format.<br>Clock format and display content                                                                                                                                                                                                                                            |
|                   | Clock content           | 1         | Clock content.<br>Clock format and display content                                                                                                                                                                                                                                           |
|                   | Text alignment          | 1         | Text alignment and line space<br>Bit0~1: the horizontal alignment (<br>0: Left-aligned<br>1: Horizontal center<br>2: Right- aligned)<br>Bit2~3: vertical alignment (<br>0:Top-aligned<br>1:Vertically center,<br>2: Bottom- aligned)<br>Bit4~7: Line space 0~15point                         |
|                   | Reserved                | 6         | Reserved, fill in 0                                                                                                                                                                                                                                                                          |
| Window            |                         | Variable- | window parameter. Each window has a                                                                                                                                                                                                                                                          |

|           |  |        |                                                                                                                                                                 |
|-----------|--|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter |  | length | 16 bytes length parameter. The total length of the data is: the number of the window*16.<br>You can see the detail at Appendix 1 : Window position and property |
|-----------|--|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Enter into or exit program template mode : CC=0x82:**

| Data Item | Value | Lenght(byte) | Description                                                                                                                                                                |
|-----------|-------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC        | 0x82  | 1            | Describe the package is the data which to enter into or exit program template mode                                                                                         |
| In or out | 1 / 0 | 1            | Bit0: Mode Action.<br>1: Enter into program template mode<br>0: Exit program template mode<br>Bit1~2: Reserved<br>Bit4: Save mode. 1 Save, 0 Not save.<br>Bit5~7: Reserved |

\* The meaning of "return value" in the return packet:

0x01 program template is invalid, can not into the program template.

**Query program template status parameter : CC=0x83:**

Send data:

| Data Item | Value | Lenght(byte) | Description                                                                                                                                                         |
|-----------|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC        | 0x83  | 1            | Describe the package is the data of query program template status parameter.                                                                                        |
| Options   | 0x00  | 1            | Bit0: Whether to query program template status parameter<br>Bit1: Whether to return the template definition color gray, screen size information<br>Bit2~7: Reserved |

Return data

| Data Item | Value | Lenght(byte) | Description                                                                         |
|-----------|-------|--------------|-------------------------------------------------------------------------------------|
| CC        | 0x83  | 1            | Describe the package is the return data of query program template status parameter. |

|                 |  |   |                                                                                                                                          |
|-----------------|--|---|------------------------------------------------------------------------------------------------------------------------------------------|
| Options         |  | 1 | The same value with send value of "Options".                                                                                             |
| Template mode   |  | 1 | 0: Not program template<br>1: program template                                                                                           |
| Template status |  | 1 | Bit0~1: template availability<br>0: the template is not available<br>1: the template can be used<br>others: Reserved<br>Bit2~7: Reserved |

**Delete program : CC=0x84:**

| Data Item      | Value | Lenght(byte)    | Description                                                                                               |
|----------------|-------|-----------------|-----------------------------------------------------------------------------------------------------------|
| CC             | 0x84  | 1               | Describe the package is the data which to delete program                                                  |
| Options        |       | 1               | Bit0: Delete program option<br>0 : Delete all program<br>1 : Delete specified program<br>others: Reserved |
| Program number |       | 1               | Delete all programs do not need this data                                                                 |
| Program list   |       | Variable-length | Length number of bytes equal to the number of programs.<br>Each program is 1 byte, program number from 1  |

\* The meaning of "return value" in the return packet:

0x01 the template is not available

0x80 Current is not a program template way

**Send text to the specified window of the specified program : CC=0x85:**

Send data

| Data Item   | Value | Lenght(byte) | Description                                                                                          |
|-------------|-------|--------------|------------------------------------------------------------------------------------------------------|
| CC          | 0x85  | 1            | Describe the package is the data which to send text to the specified window of the specified program |
| Append code |       | 4            | The user's append code, high byte previous.                                                          |
| Program No  |       | 1            | Valid value:1~100                                                                                    |
| Window No   |       | 1            | Valid value:1 ~ 10 , Invalid when out of                                                             |



|                                                                                                |                         |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------|-------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                |                         |                 | program template definition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Property                                                                                       |                         | 1               | Bit0~3: Text type<br>0 : Common Text ; 1 : Format Text<br>Bit4: Display format.<br>0: default format 1:specify format<br>Bit5~7: Reserved                                                                                                                                                                                                                                                                                                                                                                                                |
| Show format<br><br>(Note : When the "Property" display format is zero, do not need this data ) | Stay time/ Scroll times | 2               | Stay time/Scroll times: High byte first. When the show effect is scroll , it means scroll times(0 scroll one times,1 scroll two times,...) , for others , it means stay time, unit is second.                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                | Speed                   | 1               | The smaller the faster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                                | Font size               | 1               | Bit 0~3:Font size, Font size code<br>Bit 4~6:Font style , Font style code                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                                | Font color.             | 1               | Font color. Text color code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                | Show effect             | 1               | Show effect. Special effect for text and picture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                                | Text alignment          | 1               | Text alignment and line space<br>Bit0~1: the horizontal alignment ( <div style="margin-left: 20px;">                         0: Left-aligned<br/>                         1: Horizontal center<br/>                         2: Right- aligned)                     </div> Bit2~3: vertical alignment ( <div style="margin-left: 20px;">                         0:Top-aligned<br/>                         1:Vertically center,<br/>                         2: Bottom- aligned)                     </div> Bit4~7: Line space 0~15point |
|                                                                                                | Reserved                | 1               | Reserved, fill in 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Text                                                                                           |                         | Variable-length | Text data according to different types of text, the text type to see the definition of the " Property ".<br>Common Text :The text string, the end to 0x00<br>Format text: The first byte is 0x01, the followed Rich3 text, a detailed description see Formatted text data format                                                                                                                                                                                                                                                         |

## Return data

| Data Item                            | Value | Lenght(byte)    | Description                                                                                                                    |
|--------------------------------------|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| CC                                   | 0x83  | 1               | Describe the package is the return data of send text to the specified window of the specified program                          |
| Append code                          |       | 4               | The user's append code, high byte previous.                                                                                    |
| Program no                           |       | 1               | The same value with send value "Program no". Valid value: 1~100                                                                |
| Window No                            |       | 1               | The same value with send value "Window no". Valid value: 1~10, Invalid when out of program template definition.                |
| Packet loss number                   |       | 1               | The number of packets that have not yet received. Sends the first packet loss number is the total number of packets minus one. |
| The packet number of the packet loss |       | Variable-length | Packet loss packet number. Always in accordance with small to large; the first packet packet number is 0. Each package a byte. |

\*Must first send the first packet. Best to confirm the first packet sent successfully, and then send subsequent packets.

\* The meaning of "return value" in the return packet:

0x01 program template is invalid

0x11 program number is out of range

0x12 window number out of range

0x80 currently is not program template way

**Send picture to the specified window of the specified program : CC=0x86:**

## Send data

| Data Item   | Value | Lenght(byte) | Description                                                                                             |
|-------------|-------|--------------|---------------------------------------------------------------------------------------------------------|
| CC          | 0x86  | 1            | Describe the package is the data which to send picture to the specified window of the specified program |
| Append code |       | 4            | The user's append code, high byte previous.                                                             |

|                                                                                                |                         |                 |                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------|-------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program No                                                                                     |                         | 1               | Valid value:1~100                                                                                                                                                                                                                                                                  |
| Window No                                                                                      |                         | 1               | Valid value:1~10 , Invalid when out of program template definition.                                                                                                                                                                                                                |
| Picture type                                                                                   |                         | 1               | Bit0~3: Picture type<br>1 : gif image file format<br>2 : gif image file references<br>4 : simple image format, see Simple picture data format<br>Bit4: Display format.<br>0: default format 1:specify format<br>Bit5: Do you play immediately. (1:Now Playing)<br>Bit6~7: Reserved |
| Show format<br><br>(Note : When the "Property" display format is zero, do not need this data ) | Stay time/ Scroll times | 2               | Stay time/Scroll times: High byte first. When the show effect is scroll , it means scroll times(0 scroll one times,1 scroll two times,...) , for others , it means stay time, unit is second.                                                                                      |
|                                                                                                | Speed                   | 1               | The smaller the faster                                                                                                                                                                                                                                                             |
|                                                                                                | Show effect             | 1               | Show effect. Special effect for text and picture                                                                                                                                                                                                                                   |
|                                                                                                | Picture mode            | 1               | Picture mode. Picture effect code                                                                                                                                                                                                                                                  |
|                                                                                                | Reserved                | 3               | Reserved, fill in 0                                                                                                                                                                                                                                                                |
| Picture data                                                                                   |                         | Variable-length | Picture data                                                                                                                                                                                                                                                                       |

## Return data

| Data Item   | Value | Lenght(byte) | Description                                                                                              |
|-------------|-------|--------------|----------------------------------------------------------------------------------------------------------|
| CC          | 0x83  | 1            | Describe the package is the return data of send picture to the specified window of the specified program |
| Append code |       | 4            | The user's append code, high byte previous.                                                              |
| Program No  |       | 1            | The same value with send value "Program no". Valid value:1~100                                           |
| Window No   |       | 1            | The same value with send value "Window no". Valid value:1~10,Invalid                                     |

|                                      |  |                 |                                                                                                                                |
|--------------------------------------|--|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
|                                      |  |                 | when out of program template definition.                                                                                       |
| Packet loss number                   |  | 1               | The number of packets that have not yet received. Sends the first packet loss number is the total number of packets minus one. |
| The packet number of the packet loss |  | Variable-length | Packet loss packet number. Always in accordance with small to large; the first packet packet number is 0. Each package a byte. |

**\*Must first send the first packet. Best to confirm the first packet sent successfully, and then send subsequent packets.**

**\* The meaning of "return value" in the return packet:**

0x01 program template is invalid

0x11 program number is out of range

0x12 window number out of range

0x80 currently is not program template way

**Show clock/temperature in the specified window of the specified program : CC=0x87:**

Send data

| Data Item   | Value | Lenght(byte)    | Description                                                                                                                               |
|-------------|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| CC          | 0x87  | 1               | Describe the package is the data which to show clock/temperature in the specified window of the specified program                         |
| Append code |       | 4               | The user's append code, high byte previous.                                                                                               |
| Program No  |       | 1               | Valid value:1~100                                                                                                                         |
| Window No   |       | 1               | Valid value:1~10 , Invalid when out of program template definition.                                                                       |
| Type        |       | 1               | Bit0~3: Type<br>2 : Clock ; 3 : Temperature<br>Bit4: Display format.<br>0: default format 1:specify format<br>Bit5~7: Reserved, fill in 0 |
| Format      |       | Variable-length | The meaning of the attribute data according to different types                                                                            |

|  |  |  |                                                                                                                      |
|--|--|--|----------------------------------------------------------------------------------------------------------------------|
|  |  |  | Type=2 : Data see appendix 1<br>Clock/Calendar type<br>Type=3 : Data see appendix 1<br>Temperature and Humidity type |
|--|--|--|----------------------------------------------------------------------------------------------------------------------|

#### Return data

| Data Item                            | Value | Lenght(byte)    | Description                                                                                                                    |
|--------------------------------------|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| CC                                   | 0x87  | 1               | Describe the package is the return data which to show clock/temperature in the specified window of the specified program       |
| Append code                          |       | 4               | The user's append code, high byte previous.                                                                                    |
| Program No                           |       | 1               | The same value with send value "Program no". Valid value: 1~100                                                                |
| Window No                            |       | 1               | The same value with send value "Window no". Valid value: 1~10, Invalid when out of program template definition.                |
| Packet loss number                   |       | 1               | The number of packets that have not yet received. Sends the first packet loss number is the total number of packets minus one. |
| The packet number of the packet loss |       | Variable-length | Packet loss packet number. Always in accordance with small to large; the first packet packet number is 0. Each package a byte. |

#### Send alone program : CC=0x88:

#### Send data

| Data Item   | Value | Lenght(byte) | Description                                                                                                                                 |
|-------------|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| CC          | 0x88  | 1            | Describe the package is the data which to send alone program<br>Alone program is independent of the program template, it can split windows. |
| Append code |       | 4            | The user's append code, high byte previous.                                                                                                 |
| Program No  |       | 1            | Valid value: 1~100                                                                                                                          |



|                          |  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option                   |  | 1                 | Bit0~4: Reserved, fill in 0.<br>Bit5: Do you play immediately.(1 Now Playing)<br>Bit6~7: Reserved, fill in 0.                                                                                                                                                                                                                                                                                                                    |
| Reserved                 |  | 3                 | Reserved, fill in 0.                                                                                                                                                                                                                                                                                                                                                                                                             |
| Window number            |  | 1                 | Valid value 1~10                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Window information table |  | 22* Window number | Every window information table has a 22 bytes length parameter. The 1~16 bytes are window position and property, You can see the detail at Appendix 1 : Window position and property; The 17~19 bytes are window data offset; The 20~22 bytes are window data length. High byte first.<br>If no data ,then window data offset and window data length all are 0.<br>The total length of the data is: the number of the window*22. |
| Window data              |  | Variable-length   | Window play data: "Text" 、"Picture" ...<br>Byte 1 : Data Type(1 Text ; 4 Picture)<br>Byte 2 : Data Format (Like "Text type" in command 0x85 and "Picture type" in command 0x86)<br>Byte 3 : Text data or picture data.                                                                                                                                                                                                           |

## Return data

| Data Item          | Value | Lenght(byte) | Description                                                                                                                    |
|--------------------|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| CC                 | 0x88  | 1            | Describe the package is the return data which to send alone program                                                            |
| Append code        |       | 4            | The user's append code, high byte previous.                                                                                    |
| Program No         |       | 1            | Valid value:1~100                                                                                                              |
| Reserved           |       | 1            | Reserved, fill in 0.                                                                                                           |
| Packet loss number |       | 1            | The number of packets that have not yet received. Sends the first packet loss number is the total number of packets minus one. |

|                                      |  |                 |                                                                                                                                |
|--------------------------------------|--|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| The packet number of the packet loss |  | Variable-length | Packet loss packet number. Always in accordance with small to large; the first packet packet number is 0. Each package a byte. |
|--------------------------------------|--|-----------------|--------------------------------------------------------------------------------------------------------------------------------|

**\*Must first send the first packet. Best to confirm the first packet sent successfully, and then send subsequent packets.**

**\* The meaning of "return value" in the return packet:**

0x01 program template is invalid

0x11 program number is out of range

0x12 window number out of range

0x13 The definition of the window outside the screen size of the program template definition

0x80 currently is not program template way

### **Query program information : CC=0x89:**

Send data:

| Data Item  | Value | Lenght(byte) | Description                                                                                                                                                   |
|------------|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC         | 0x89  | 1            | Describe the package is the data which to query program info                                                                                                  |
| Info flag  |       | 1            | Special which program info will to be query<br>1: Query valid programs count and program number<br>2: Query specifies program information.<br>Other: Reserved |
| parameters |       | 5            | The meaning of the parameters according to the info flag different.                                                                                           |

### **Parameter and return data description**

- Query "valid program count and program number"

Parameter

|          |                  |
|----------|------------------|
| Byte 1~5 | Reserved, fill 0 |
|----------|------------------|

Return data

| Data Item | Value | Lenght(byte) | Description                                                          |
|-----------|-------|--------------|----------------------------------------------------------------------|
| CC        | 0x89  | 1            | Describe the package is the return data packet of query program info |
| Info flag |       | 1            | Same with send value "info flag"                                     |

|                      |  |                 |                                                                 |
|----------------------|--|-----------------|-----------------------------------------------------------------|
| parameters           |  | 5               | Same with send value "parameters"                               |
| Valid program count  |  | 1               | Valid program count                                             |
| Valid program number |  | Variable-length | Each byte identifies an effective program. Valid value 1 ~ 100. |

\* The meaning of "return value" in the return packet:

0x01 Controller not running in program template mode

0x10 Unknown info flag

● **Query specifies program information**

Parameter

|          |                  |
|----------|------------------|
| Byte 1   | Program number   |
| Byte 2~5 | Reserved, fill 0 |

Return data

| Data Item         | Value | Lenght(byte) | Description                                                          |
|-------------------|-------|--------------|----------------------------------------------------------------------|
| CC                | 0x89  | 1            | Describe the package is the return data packet of query program info |
| Info flag         |       | 1            | Same with send value "info flag"                                     |
| parameters        |       | 5            | Same with send value "parameters"                                    |
| Information count |       | 1            | Now only return one information                                      |
| Program number    |       | 1            | Program number                                                       |
| User append code  |       | 4            | User append code                                                     |

\* The meaning of "return value" in the return packet:

0x01 Controller not running in program template mode

0x10 Unknown info flag

0x11 Invalid programs

0x12 Can't get program information

**Set program property : CC=0x8a:**

| Data Item | Value | Lenght(byte) | Description                                                    |
|-----------|-------|--------------|----------------------------------------------------------------|
| CC        | 0x8a  | 1            | Describe the package is the data which to set program property |
| Option    |       | 1            | Bit0: Set the range of the program                             |

|                 |  |                 |                                                                                                                                                                                                                                                          |
|-----------------|--|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |  |                 | property<br>0: All programs<br>1: Specify program<br>Other: Reserved                                                                                                                                                                                     |
| Program number  |  | 0/1             | The count of the program. When it sets all program property , don't need this data.                                                                                                                                                                      |
| Program list    |  | Variable-length | The list of the program. When it sets all program property , don't need this data. Length number of bytes equal to the number of programs. Each program is 1 byte, program number from 1                                                                 |
| Property flag 1 |  | 1               | marked which property you want to set by byte, set 0 if the data not exist.<br>Bit0: The level of the program.<br>Bit1: The cycle count.<br>Bit2: Valid time. How long will the program be valid from now on.<br>Bit3: Interval time<br>Bit4~7: Reserved |
| Property flag 2 |  | 1               | Bit0~4: valid time. >0 the count of the valid time.<=4<br>Bit5~7: Reserved                                                                                                                                                                               |
| Program level   |  | 1               | The program level. 1~3 level, The high level of the program is priority.                                                                                                                                                                                 |
| Play loop count |  | 2               | Loop count, High byte previous(big-endian).<br>0: Do not play the program, use to shield program temporarily.<br>1~255: The loop count of the program.                                                                                                   |
| Valid time      |  | 2               | High byte previous (big-endian). In minute.                                                                                                                                                                                                              |

|                  |  |                        |                                                                                                                                                                       |
|------------------|--|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |  |                        | 0: Not limit play time<br>>0: Specify play time in minute.                                                                                                            |
| Expiration date  |  | 12                     | Validity start date: "Year Month Day Hour Minute Second"<br>The expiry date: "Year Month Day Hour Minute Second"<br>"Year Month Day Hour Minute Second" each one byte |
| Effective period |  | 6*<br>period<br>number | Period start: "Hour Minute Second"<br>Period end: "Hour Minute Second"<br>"Hour Minute Second" each one byte                                                          |

The meaning of "return value" in the return packet:

0x01 program template is invalid

0x80 currently is not program template way

### **Set play plan : CC=0x8b:**

| Data Item        | Value | Lenght(byte) | Description                                                                                                                                                               |
|------------------|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC               | 0x8b  | 1            | Describe the package is the data which to set play plan                                                                                                                   |
| Append code      |       | 4            | The user's append code, high byte previous.                                                                                                                               |
| Plan No          |       | 1            | Valid value 1~100. Total support 100 plans, For each plan No, the new data cover the old data                                                                             |
| Format and level |       | 1            | Bit0~3: Data format, fill in 0x01<br>Bit4~7: Indicates the priority level. The priority level the greater the value, the more priority to play, 0 is the lowest priority. |
| Weekday          |       | 1            | Bit0~6: 7-bit logo Sunday to Saturday                                                                                                                                     |
| Begin date       |       | 3            | 3 bytes:<br>Byte1:Year,Valid value0~99,means 2000~2999<br>Byte2:Month<br>Byte3:Day                                                                                        |
| End date         |       | 3            | 3 bytes:<br>Byte1:Year,Valid value0~99,means 2000~2999                                                                                                                    |



|                |  |                 |                                                                                 |
|----------------|--|-----------------|---------------------------------------------------------------------------------|
|                |  |                 | Byte2:Month<br>Byte3:Day                                                        |
| Begin time     |  | 3               | 3 bytes:<br>Byte1:Hour<br>Byte2:Minute<br>Byte3:Second                          |
| End time       |  | 3               | 3 bytes:<br>Byte1:Hour<br>Byte2:Minute<br>Byte3:Second                          |
| Program number |  | 1               | Valid value:1 ~100                                                              |
| Program No     |  | Variable-length | Each byte represents a program. Numbered in ascending order, do not repeat....。 |

Total support 100 plans, For each plan No, the new data cover the old data .

Return data

| Data Item   | Value | Lenght(byte) | Description                                                                                   |
|-------------|-------|--------------|-----------------------------------------------------------------------------------------------|
| CC          | 0x8b  | 1            | Describe the package is the return data which to set play plan                                |
| Append code |       | 4            | The user's append code, high byte previous.                                                   |
| Plan No     |       | 1            | Valid value 1~100。 Total support 100 plans, For each plan No, the new data cover the old data |

The meaning of "return value" in the return packet:

0x01 program template is invalid

0x80 currently is not program template way

### Delete play plan : CC=0x8c:

| Data Item        | Value | Lenght(byte) | Description                                                               |
|------------------|-------|--------------|---------------------------------------------------------------------------|
| CC               | 0x8c  | 1            | Describe the package is the data which to delete play plan                |
| Append code      |       | 4            | The user's append code, high byte previous.                               |
| Play plan number |       | 1            | The number of play plan will to be delete。 0 means delete all play plans. |
| Play plan        |       | Variable-    | Valid value 1~100。 Each byte                                              |

|    |  |        |                                                                                                          |
|----|--|--------|----------------------------------------------------------------------------------------------------------|
| No |  | length | represents a play plan no.<br>When delete all play plan, the length of this data is one , value is 0xff. |
|----|--|--------|----------------------------------------------------------------------------------------------------------|

Total support 100 plans.

When delete all play plans, the play plan number fill in 0, the length of play plan no is one , value is 0xff.

Return data

| Data Item   | Value | Lenght(byte) | Description                                                                                   |
|-------------|-------|--------------|-----------------------------------------------------------------------------------------------|
| CC          | 0x8c  | 1            | Describe the package is the return data which to delete play plan                             |
| Append code |       | 4            | The user's append code, high byte previous.                                                   |
| Plan No     |       | 1            | Valid value 1~100. Total support 100 plans, For each plan No, the new data cover the old data |

The meaning of "return value" in the return packet:

0x01 program template is invalid

0x11 The number of play plan will to be delete is 0.

0x80 currently is not program template way

### **Query play plan : CC=0x8d:**

Send data

| Data Item    | Value | Lenght(byte) | Description                                                                         |
|--------------|-------|--------------|-------------------------------------------------------------------------------------|
| CC           | 0x8d  | 1            | Describe the package is the data which to query play plan                           |
| Append code  |       | 4            | The user's append code, high byte previous.                                         |
| Query type   |       | 1            | 0: Query all valid play plan.<br>1: Query specified play plan no<br>Other: Reserved |
| Play plan No |       | 1            | Valid value:1~100.<br>When query type is 0, this data fill in 0.                    |

Total support 100 plans.

Return data

| Data Item | Value | Lenght(byte) | Description                                                      |
|-----------|-------|--------------|------------------------------------------------------------------|
| CC        | 0x8d  | 1            | Describe the package is the return data which to query play plan |

|                                                      |  |                 |                                                                                                                                                                   |
|------------------------------------------------------|--|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Append code                                          |  | 4               | The user's append code, high byte previous.                                                                                                                       |
| Query type                                           |  | 1               | 0: Query all valid play plan.<br>1: Query specified play plan no<br>Other: Reserved                                                                               |
| Count/Number                                         |  | 1               | When query type is 0, this value is valid play schedule count<br>When query type is 1, this value is play schedule number.                                        |
| Play schedule number table/<br>play schedule content |  | Variable-length | When query type is 0, this value is valid play schedule number table<br>When query type is 1, this value is play schedule content. Data format like command 0x8B. |

You must deal with the return data according to the different query type.

The meaning of "return value" in the return packet:

0x01 program template is invalid

0x11 Don't support the query type.

0x12 Invalid play plan no.

0x80 currently is not program template way

## Appendix 1: Window position and property

|      |   |   |   |   |    |   |    |   |                 |   |   |   |   |   |   |   |
|------|---|---|---|---|----|---|----|---|-----------------|---|---|---|---|---|---|---|
|      | 0 | 1 | 2 | 3 | 4  | 5 | 6  | 7 | 8               | 9 | A | B | C | D | E | F |
| 0x00 | x |   | y |   | cx |   | cy |   | Window property |   |   |   |   |   |   |   |

Description :

| Data Item       | Data Size(BYTE) | Description                                            |
|-----------------|-----------------|--------------------------------------------------------|
| x               | 2               | Window start point x, high byte previous (big endian). |
| y               | 2               | Window start point Y, high byte previous (big endian). |
| cx              | 2               | Window width. High byte previous (big endian).         |
| cy              | 2               | Window height. High byte previous (big endian).        |
| Window property | 8               | Window default type and parameters.                    |

Window property is represent by 8 bytes :

|      |      |           |   |   |   |   |   |   |
|------|------|-----------|---|---|---|---|---|---|
|      | 0    | 1         | 2 | 3 | 4 | 5 | 6 | 7 |
| 0x00 | mode | Parameter |   |   |   |   |   |   |

The definition of the window mode :

| Mode value | Description                      |
|------------|----------------------------------|
| 0          | Blank(Show nothing)              |
| 1          | Text                             |
| 2          | Clock、calendar                   |
| 3          | Temperature、Humidity             |
| 4          | Picture、Reference of the picture |
| Other      | Reserved                         |

The parameter has different values according to the mode. There are all the mode's parameters. All reserved position should be set 0x00.

### **Blank type**

|      |   |          |   |   |   |   |   |   |
|------|---|----------|---|---|---|---|---|---|
|      | 0 | 1        | 2 | 3 | 4 | 5 | 6 | 7 |
| 0x00 | 0 | Reserved |   |   |   |   |   |   |

### **Text type**

|      |   |        |           |            |       |           |       |   |
|------|---|--------|-----------|------------|-------|-----------|-------|---|
|      | 0 | 1      | 2         | 3          | 4     | 5         | 6     | 7 |
| 0x00 | 1 | Effect | Font size | Font color | Speed | Stay time | Align |   |

| Data Item                  | Value         | Length(BYTE) | Description                                                                                                  |
|----------------------------|---------------|--------------|--------------------------------------------------------------------------------------------------------------|
| Effect                     | 1             | 1            | Effect: See in Special effect for text and picture                                                           |
| Font size                  |               | 1            | Bit0~2: Font size, see in Font size code<br>Bit 4~6:Font style , Font style code                             |
| Font color                 |               | 1            | Bit0~2: Font color , see 1-byte color value in Text color code                                               |
| Speed                      | 0~9           | 1            | The smaller the faster                                                                                       |
| Stay time/<br>Scroll times | 0x0000~0xffff | 2            | Stay time/Scroll times: High byte first. When the show effect is scroll , it means scroll times(0 scroll one |

|       |  |   |                                                                                                                                                                                                                                                                      |
|-------|--|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |  |   | times,1 scroll two times,...) , for others , it means stay time, unit is second.                                                                                                                                                                                     |
| Align |  | 1 | Text alignment and line space<br>Bit0~1: the horizontal alignment (<br>0: Left-aligned<br>1: Horizontal center<br>2: Right- aligned)<br>Bit2~3: vertical alignment (<br>0:Top-aligned<br>1:Vertically center,<br>2: Bottom- aligned)<br>Bit4~7: Line space 0~15point |

### *Clock/Calendar type*

|      |   |           |            |           |          |        |         |   |
|------|---|-----------|------------|-----------|----------|--------|---------|---|
|      | 0 | 1         | 2          | 3         | 4        | 5      | 6       | 7 |
| 0x00 | 2 | Font size | Font color | Stay time | calendar | Format | Content |   |

| Data Item  | Value         | Length(BYTE) | Description                                                                      |
|------------|---------------|--------------|----------------------------------------------------------------------------------|
| Font size  |               | 1            | Bit0~2: Font size, see in Font size code<br>Bit 4~6:Font style , Font style code |
| Font color |               | 1            | Bit0~2: Font color , see 1-byte color value in Text color code                   |
| Stay time  | 0x0000~0xffff | 2            | High byte previous(big endian), unit is second.                                  |
| Calendar   |               | 1            | 0: The gregorian calendar                                                        |
| Format     |               | 1            | Clock format: See in Clock format and display content                            |
| Content    |               | 1            | Clock content: see in Clock format and display content                           |



### Temperature and Humidity type

|      |   |           |            |           |        |          |   |   |
|------|---|-----------|------------|-----------|--------|----------|---|---|
|      | 0 | 1         | 2          | 3         | 4      | 5        | 6 | 7 |
| 0x00 | 3 | Font size | Font color | Stay time | Format | Reserved |   |   |

| Data Item  | Value         | Length(BYTE) | Description                                                                      |
|------------|---------------|--------------|----------------------------------------------------------------------------------|
| Font size  |               | 1            | Bit0~2: Font size, see in Font size code<br>Bit 4~6:Font style , Font style code |
| Font color |               |              | Bit0~2: Font color , see 1-byte color value in Text color code                   |
| Stay time  | 0x0000~0xffff | 2            | High byte previous(big endian ), unit is second.                                 |
| Format     |               | 1            | 0: Celsius<br>1: Fahrenheit<br>2: Humidity                                       |

### Picture and reference to the picture

|      |   |        |       |           |          |   |   |   |
|------|---|--------|-------|-----------|----------|---|---|---|
|      | 0 | 1      | 2     | 3         | 4        | 5 | 6 | 7 |
| 0x00 | 4 | Effect | Speed | Stay time | Reserved |   |   |   |

| Data Item | Value         | Length(BYTE) | Description                                                                                                                                                                                   |
|-----------|---------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effect    |               | 1            | Effect: See in Special effect for text and picture                                                                                                                                            |
| Speed     | 0~9           | 1            | The smaller of the value, the faster. Invalid when display immediately.                                                                                                                       |
| Stay time | 0x0000~0xffff | 2            | Stay time/Scroll times: High byte first. When the show effect is scroll , it means scroll times(0 scroll one times,1 scroll two times,...) , for others , it means stay time, unit is second. |

